



MLOPS COMMUNITY · RESEARCH COPILOT WORKSHOP

Making Scientific Knowledge *Navigable* for Agents



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Scientific knowledge is fragmented.

We query it as if it were flat text — and we can do better.



Knowledge is scattered

Literature, datasets, software, posters, talks — in different systems, with different identifiers, and no common seams.



The structure we have is incomplete

DOIs, citation graphs, data-availability statements, ORCIDs — partial coverage, inconsistent metadata, ambiguous attribution.



Relationships are underspecified

A citation can mean: builds on, disproves, mentions, reuses data from, or reuses methods from. The graph collapses all five into one edge.



Search is the universal API

Keyword or vector search returns a ranked list. Fine for 'find me a paper.' Not enough for 'map the landscape' or 'what breaks if this is wrong.'

Research output accelerates like codebases do.

Engineers gave their agents structure. Scientists haven't — yet.

5M+

papers per year

doubling every ~12 years

~25%

of researcher time

spent on literature search

AI×

acceleration

same curve hits code and research

Codebases outran human reading years ago. Engineers responded with dependency graphs, call graphs, blast-radius tools. Scientific literature has the same shape of problem — and hasn't yet had the same response.

Fragmentation: serendipity over systems.

Right now, transferable methods travel by chance. That's the thing to fix.



Citations encode more than dependency

Prestige, convention, self-citation, availability bias. A clean technical DAG is not what the citation graph actually is.



Disciplinary silos hide transferable methods

A technique useful in one field may have been solved in a neighbouring one years earlier — under different vocabulary.



Connections happen through serendipity

Whether two lines of work meet depends on who read what, when. It is not a structural property of the knowledge itself.



From serendipity
to retrievable
relevance.

*So an agent can surface the connection,
and a researcher can rely on it.*

Codebases as analogy.

Software stopped treating code as isolated files. Science can do the same.

SOFTWARE ENGINEERING

Dependencies

what code relies on to run

Call graphs

chains of function calls

Blast radius

what breaks if this changes

SCIENCE



Reused methods and datasets

what a result relies on to hold



Chains of reasoning

a claim resting on a claim resting on a dataset



Result impact

which follow-on findings are at risk

Citation graphs approximate all of this — but they're incomplete, semantically weak, and socially biased. The real signal is the underlying ecosystem of data, methods, results, code, and claims.

Toward 'debugging' science.

Treat outputs as claims to evaluate — independent of citation count or venue prestige.

IMPLICIT ACCEPTANCE CRITERIA

Science already has these.

- **Physical consistency**

does it fit known laws?

- **Reproducibility**

can someone else hit the same result?

- **Compatibility**

does it agree with adjacent findings?

WHAT A NAVIGABLE LAYER UNLOCKS



Automated consistency checks

Flag claims that contradict well-established adjacent results.



Conflicting-result detection

Surface studies that disagree rather than burying the tension.



Missing-link identification

Papers that should cite each other but don't — and why.



Method-reuse surfacing

Transferable techniques across silos, beyond nearest-neighbor prose.

What 'navigable' means for an agent.

Agents benefit from ranked lists too — they just benefit from more.

HUMAN SEARCHER

Ranked list

Scans titles, opens a few.

One-shot query

Session state lives in the researcher's head.

Visual cues

Journal, authors, year, formatting.

Stops when tired

Implicit budget, intuition-driven.

AGENT — GIVEN THE RIGHT PRIMITIVES

Graph topology

Which papers bridge communities? Which are central?

Session working set

Accumulates, dedupes, tags, reasons across turns.

Structured returns

Bibcodes, IDs, community labels, provenance.

Multi-hop moves

Citation chains, co-citation, bridge-paper gaps, temporal.

Vector search and knowledge graphs, together.

Hybrid retrieval is table stakes. Agent-mediated iteration is the unlock.



Semantic

Dense vectors — domain-tuned plus general-purpose, fused via reciprocal rank.



Lexical

BM25 on titles, abstracts, keywords. Catches specific terms that embeddings blur.



Structural

Graph moves: citation chains, co-citation, shortest paths, community lenses.



Symbolic

Entity and claim extraction — instruments, datasets, methods, results as first-class objects.



Iterative

An agent chooses the next move based on what it just learned. The loop — not any single tool — is where the leverage lives.

Why MCP is a good fit for this layer.

Model Context Protocol gives the agent tools it can actually hold.



Tool-calling primitives

First-class: name, typed args, typed returns. The agent knows what it can do.



Session context

Stateful per-connection. Working sets live on the server, not in the prompt.



Structured responses

JSON objects, not HTML snippets. Compose cleanly with other tools.



Transport-agnostic

stdio, HTTP, WebSocket. Works with Claude Desktop, Cursor, custom agents.

PORTABLE DISCIPLINE

If your research copilot speaks MCP, the same knowledge layer can be hosted by ADS, arXiv, your own group, or someone else.

A WORKING PROTOTYPE

MCP for NASA ADS/SciX.

PostgreSQL + pgvector. Single server. One MCP endpoint.

32.4M

papers indexed

Deliberate simplicity: no separate vector DB, no orchestrator, no queue. Postgres + pgvector carries the corpus, the graph metrics, and the retrieval stack.

STACK

PostgreSQL

corpus + graph metrics

pgvector

HNSW dense ANN

Qdrant

rec subset w/ payloads

MCP server

single endpoint

Two vector stores, two jobs.

Pick the backend for the move — the agent only sees one MCP interface.

pgvector

THE CORPUS

- All 32.4M papers
- Dense ANN via HNSW, joined with graph metrics and communities
- One SQL query composes retrieval + filters + PageRank

Good at: topical search, hybrid RRF, structural filters.

Qdrant

THE RECOMMENDER

- Pilot subset (top-N by PageRank) with rich payload
- Multi-example recommendation: 'more like these, less like those'
- Payload filters: year, doctype, community, arxiv class

Good at: what pgvector can't cleanly do in SQL.

One MCP tool surface (`find_similar_by_examples`). Two backends. The agent picks the move by name — not by where the vectors live.

Principles for agent navigation.

From watching agents actually use these tools.

01

Fewer, composable tools > many specific tools

Agent selection accuracy degrades past a point. Prefer a small set of primitives the agent can combine.

02

Return compact stubs, full text on request

bibcode · title · first_author · year · citation_count · snippet. Context is the scarcest resource.

03

Stable, typed schemas

JSON, no free-text fields to parse. Every response composes cleanly with whatever the agent does next.

04

Include provenance

Which tool returned this? Which query? The agent can reason about its own trail — and you can, in the trace.

05

Cursor-based pagination

Never return 10,000 bibcodes in one payload. Give the agent a cursor and let it ask for more.

06

Idempotent reads, explicit side effects

Same inputs → same outputs. Anything that writes state is called out in the tool name.

Exploration modes beyond search.

Each of these is a move an agent takes — not a UI the human drives.



Bridge papers between communities

'Which papers are cited by both exoplanet and atmospheric-chemistry communities?' as a single structural query.



Citation chains

Follow the chain of results this paper rests on. Where does it terminate — a dataset? a theorem? an assumption someone flagged?



Co-citation & bibliographic coupling

Papers that are read-alike (cited together) or reference-alike (share references). Invisible-college detectors.



Temporal evolution

Which subfields are growing? Which are consolidating into review papers? Which had a flurry five years ago and went quiet?



Multi-lens community assignment

Three lenses — citation, semantic, taxonomic. Their disagreements are informative: semantically central but citationally isolated is often an 'outsider with a good idea.'

Three community lenses.

Each has clean semantics. The agent picks the one that fits the question.

Citation

BEHAVIORAL AFFINITY

- Leiden on full citation graph
- 3 resolutions: $\sim 20 \cdot \sim 200 \cdot \sim 2000$
- Coverage: 62% of papers

'Papers that cite each other' — finds invisible colleges

Semantic

TOPICAL AFFINITY

- k-means on INDUS embeddings
- 3 resolutions
- Coverage: 99% (all embedded)

'Papers with similar content' — catches cross-citing subfields

Taxonomic

INSTITUTIONAL AFFINITY

- arXiv class + UAT concepts
- Hierarchical, human-curated
- Coverage: $\sim 8\%$ (arXiv subset)

'Papers in the same field' — sanity-check for the others

Semantically central but citationally isolated is usually an outsider with a good idea — or a review the field hasn't caught up to.

EXAMPLE

Mapping the exoplanet-atmospheres landscape.

USER PROMPT

"I'm preparing a grant on exoplanet atmospheric characterization. Map the research landscape — major subcommunities, foundational papers, bridges between communities, and what's worth reading first."

MOVES THE COPILOT MAKES

1

search

broad hybrid hit on the topic

2

graph_context

communities + PageRank for top hits

3

citation_similarity

expand with co-citation bridges

4

temporal_evolution

see which communities are growing

5

find_gaps

bridge papers between communities

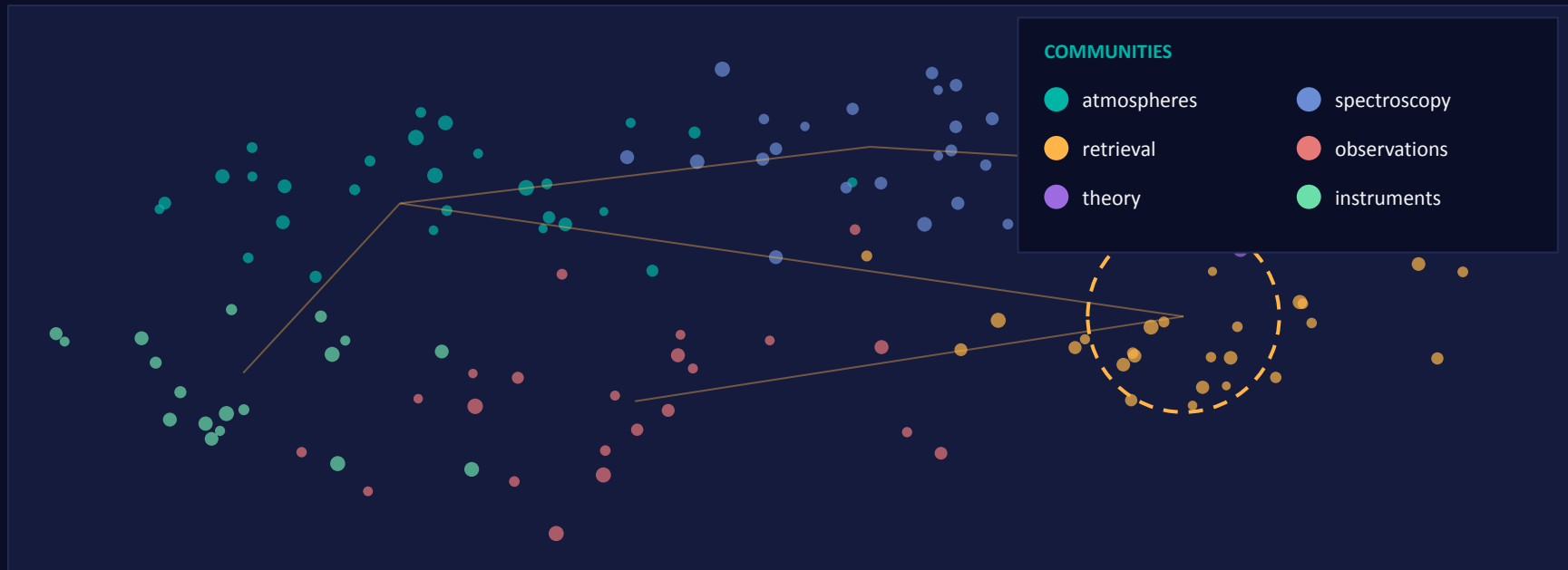
6

read_paper

deep-dive on 3 foundational candidates

What the copilot sees while it works.

Not visually — as graph edges, community IDs, and embedding coordinates.



Using a research copilot — the questions to ask.

Most of you will be integrating one into your workflows, not necessarily building the knowledge layer.

1

What does the corpus cover?

Full field or a rolling window? Do the graph metrics it returns mean what you think they mean?

2

What navigation does it support beyond hybrid search?

Citation chains, co-citation, community bridges, temporal slices. If the only primitive is search, you have a faster librarian.

3

Does it return structured objects?

A unique identifier, not a snippet. IDs compose; prose doesn't. Prose forces the agent to parse its own context.

4

Can you trace what it consulted?

Every answer should come with the tool calls it made. This is how you trust, debug, and teach it.

5

Does it complement your judgement?

Use it to expand the landscape. Reserve peer review, interpretation, and prioritization for the human.

TAKEAWAY

SEARCH → *UNDERSTANDING*

RETRIEVAL → *NAVIGABLE CONTEXT*

SERENDIPITY → *SYSTEMS*

Scientific knowledge gets more useful when we treat it like the system it already is — data, methods, results, code, claims — and let agents navigate the connections we've been leaving to chance.

THANK YOU

Questions?

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RESOURCES



Repository

github.com/sjarmak/scix-agent



Live MCP

Coming soon on <https://scixplorer.org>



Paper

ADASS 2026 · arXiv preprint in prep



Contact

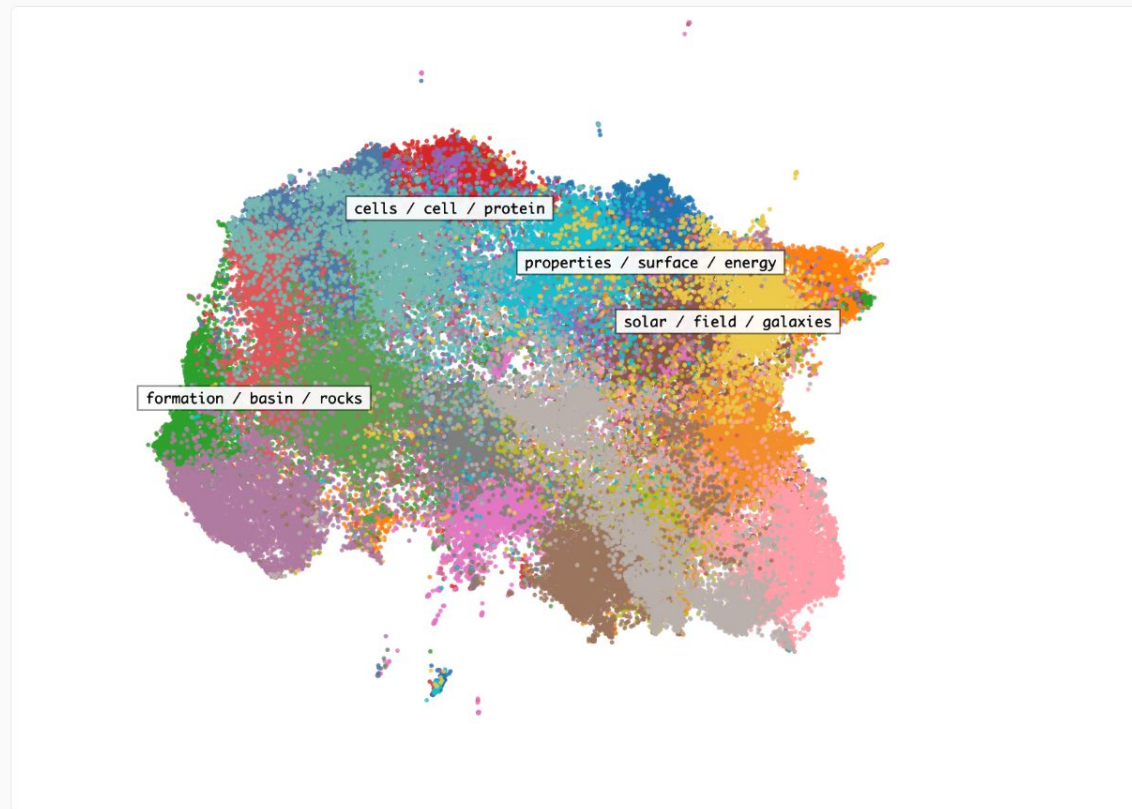
stephanie.jarmak@sourcegraph.com

Bonus slides

UMAP Embedding Browser

100,000 papers · 20 communities

2-D UMAP projection of INDUS paper embeddings. Pan and zoom to explore; hover a point to see its title; click to pin a paper. Click a legend switch to isolate a community.



Selected paper

bibcode: 1971ZPhy...243...18**community:** 86

Inelastic collision cross section of excited molecules

[Show neighborhood](#)[Reset colors](#)[Open citation ego network →](#)

Communities (coarse)

Click to isolate · click again to reset

- | | |
|---|--|
| properties / surface / energy (5,000) | solar / field / galaxies (5,000) |
| formation / basin / rocks (5,000) | cells / cell / protein (5,000) |
| water / flow / temperature (5,000) | temperature / surface / properties (5,000) |
| project / research / students (5,000) | research / development / students (5,000) |
| control / algorithm / design (5,000) | water / chemical / process (5,000) |
| species / soil / plant (5,000) | performance / frequency / design (5,000) |
| sea / ocean / water (5,000) | soil / water / species (5,000) |
| water / climate / precipitation (5,000) | quantum / energy / field (5,000) |
| seismic / earthquake / flow (5,000) | power / control / voltage (5,000) |
| image / algorithm / learning (5,000) | energy / power / performance (5,000) |

Agent Trace Overlay

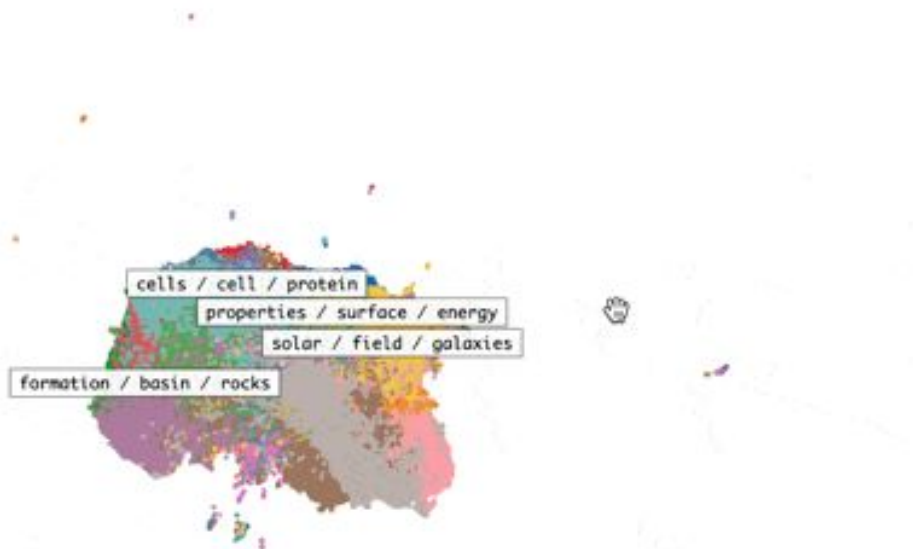
Type a query below to run a real MCP tool sequence against the 32M-paper corpus. Each tool call narrates in the panel and the touched papers flash on the UMAP overlay. Preset pills swap between the single-tool search and the multi-step scenarios.

e.g. JWST asteroid

Search

Search · JWST asteroid

Lit Review · galaxy clusters dark matter



Agent trace

200 events

clear

SEARCH: Single hybrid search — INDUS dense embedding + BM25 lexical, fused via RRF. One trace event with the top-N result set.

- 19:11:41 search 5 bibcodes 5554.7 ms
- 19:11:45 graph_context 1 bibcode 3588.1 ms
- 19:11:45 graph_context 0 bibcodes 5.8 ms
- 19:11:45 graph_context 0 bibcodes 3.1 ms
- 19:11:45 temporal_evolution 0 bibcodes 21.7 ms
- 19:11:45 get_paper 1 bibcode 44.2 ms
- 19:12:14 search 5 bibcodes 28979.8 ms
- 19:12:14 citation_similarity 0 bibcodes 13.9 ms
- 19:12:27 concept_search 5 bibcodes 13881.5 ms
- 19:12:31 concept_search 5 bibcodes 4255.7 ms
- 19:12:31 get_paper 1 bibcode 2.4 ms
- 19:20:59 lit_review 5 bibcodes 13871.6 ms
- 19:21:19 search 5 bibcodes 6473.2 ms
- 19:21:19 get_paper 1 bibcode 23.6 ms
- 19:21:20 get_paper 1 bibcode 33.6 ms
- 19:21:20 get_paper 1 bibcode 17.2 ms
- 19:21:21 lit_review 5 bibcodes 1378.9 ms
- 19:21:21 entity 0 bibcodes 199.6 ms
- 19:21:21 entity 0 bibcodes 3.3 ms
- 19:21:21 entity_context 0 bibcodes 1.8 ms

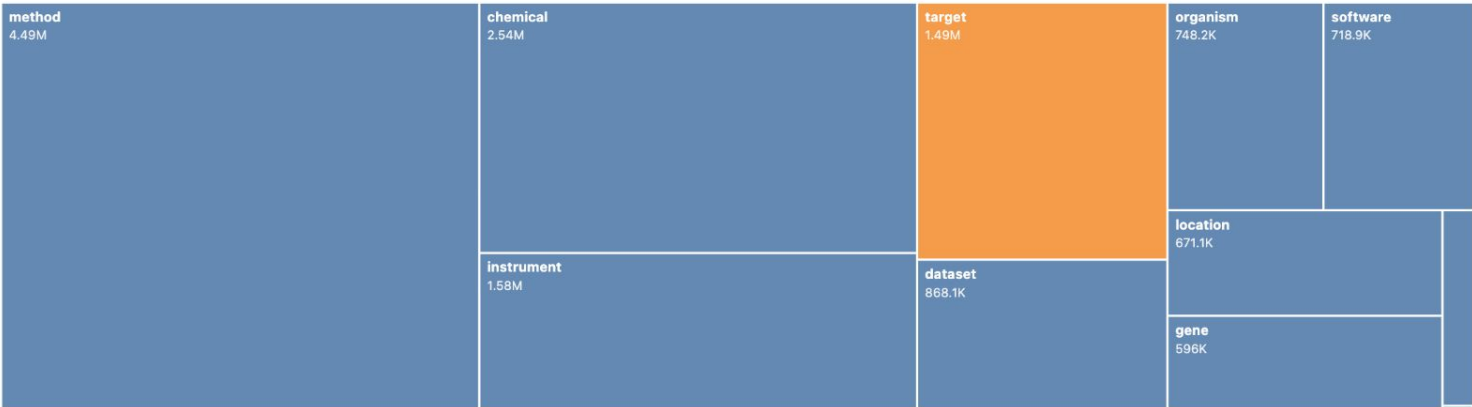
Entity Coverage Dashboard

What does ADS know about? 13 entity vocabularies harvested from 12 sources, ~13.5M canonical entities, ~70M paper↔entity links. Most lit-search systems have one or two entity types; SciX has taxonomic, planetary-body, and instrument-class data that no general literature engine carries. The treemap shows vocabulary size; the stacked bar shows source provenance; the coverage-gap callout flags vocabularies still in their early-harvest phase.

13.84M canonical entities 13 vocabularies 12 sources 71.41M paper↔entity links as of 2026-04-27T22:50:58Z

Vocabularies

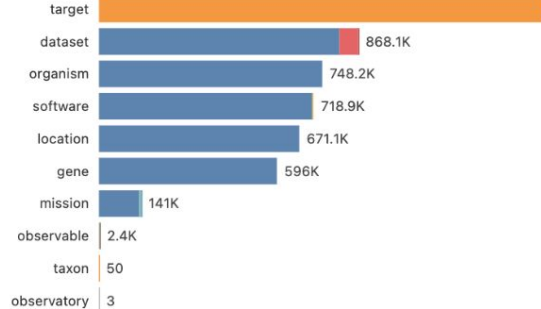
Treemap area is proportional to entity count; tile color encodes the dominant source. Hover for full numbers.



Source provenance per vocabulary

Stacked bars showing which sources contribute to each vocabulary. ``gliner`` (the LLM NER pipeline) dominates the high- volume vocabularies; ``ssodnet`` is the planetary-body specialist; the 9 smaller curated sources cover instruments, datasets, codes, and physics keywords.





Coverage gaps

Vocabularies with fewer than 10K entities — early-harvest stories that show where SciX is still growing.

observable

2,424 entities · 7,074,541 links

taxon

50 entities · 0 links

observatory

3 entities · 1,965 links

Top-N most-linked entities

The entities with the most paper links per vocabulary. Pick a vocabulary on the left to see which canonical names appear most frequently in the document graph.

method	27.29M links
chemical	11.73M links
instrument	7.63M links
target	17.1K links
dataset	2.1M links
organism	3.96M links
software	1.75M links
location	2.85M links
gene	1.98M links
mission	5.02M links
observable	7.07M links
taxon	0 links

#	name	source	ambiguity	links
1	Iris	ssodnet	banned	4,045
2	Vesta	ssodnet	unique	3,318
3	Pallas	ssodnet	domain_safe	2,090
4	Ryugu	ssodnet	unique	1,376
5	Bennu	ssodnet	unique	1,340
6	Itokawa	ssodnet	domain_safe	965
7	Metis	ssodnet	unique	924
8	Flora	ssodnet	banned	472
9	Ceres	ssodnet	homograph	451
10	Lutetia	ssodnet	domain_safe	422
11	Gaspra	ssodnet	domain_safe	318

within-community citations; off-diagonal = cross-discipline flow.

20×20 coarse
 200×200 medium (dense overview, 20 coarse domains)
 raw counts
 row-normalized (per source)
 citation lift (vs uniform)

1 41,297,573

Source community (citing)

Target community (cited)



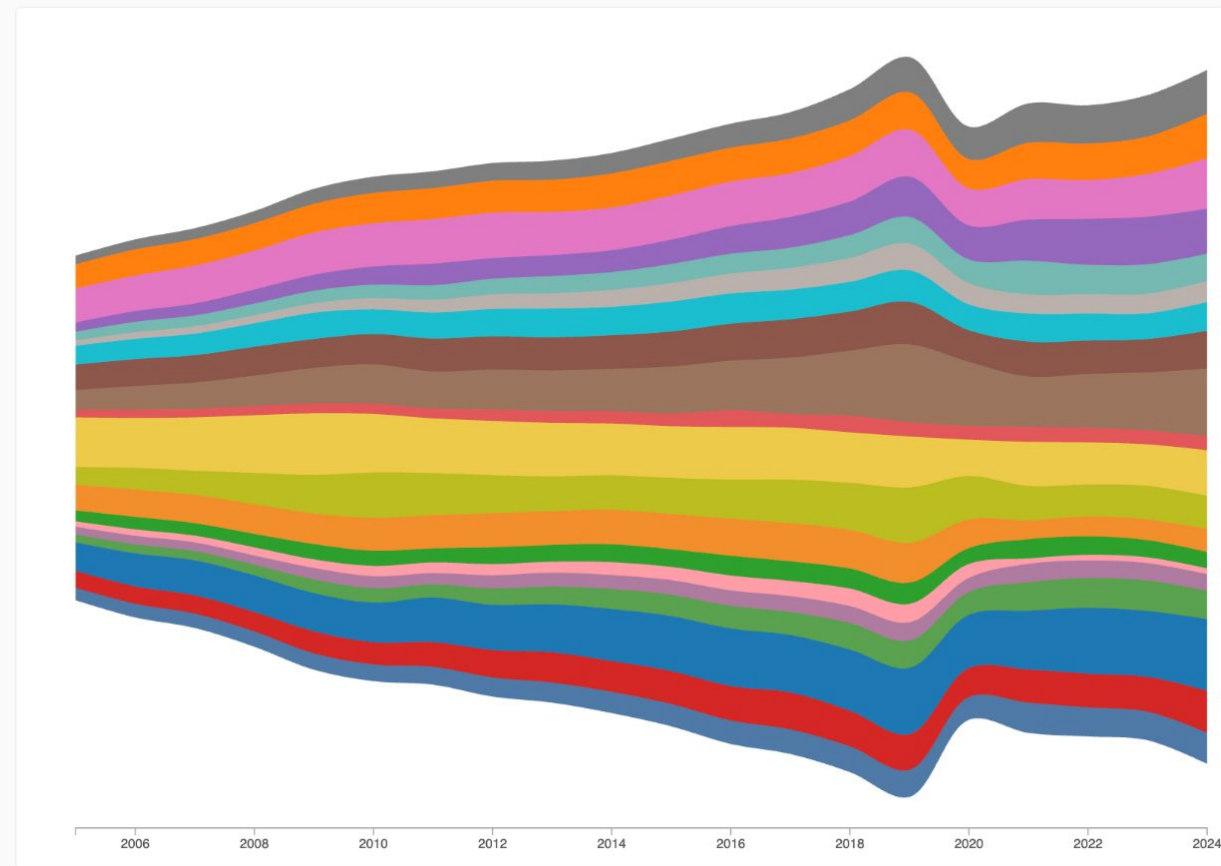
Click a cell

Hover a cell for exact count. Click to pin source/target communities and their top terms here.

Field Growth (Streamgraph)

19,605,201 papers · 20 communities · 2005–2024

Papers per community per year, stacked with a streamgraph offset. Width of each band reflects publication volume; vertical position is cosmetic. Hover a band for the community label and per-year counts. Switch the resolution toggle in the nav to flip between ~20 broad domains and ~200 fine-grained topics.



Top 12 communities

By total papers in the visible window. Hover a band for details.

properties / surface / energy	1,781,503
quantum / energy / field	1,777,843
image / algorithm / learning	1,624,702
project / research / students	1,481,059
control / algorithm / design	1,276,103
temperature / surface / properties	1,165,104
solar / field / galaxies	1,147,379
performance / frequency / design	1,071,537
cells / cell / protein	1,014,645
water / flow / temperature	938,745
water / chemical / process	935,195
research / development / students	845,229

Citation Ego Network

Force-directed view of a paper's citation neighborhood. Blue edges are papers the center cites; orange edges are papers that cite the center; grey edges reach 2-hop neighbors. Drag nodes to rearrange. Click a neighbor to re-center on it. Hover for bibcode + title.

42 refs · 11 cites · 200 2-hop · 942 edges · 4649 ms

2021ApJ...910..107G

Load



Center paper

bibcode: 2021ApJ...910..107G

community: —

Backstreaming Pickup Ions

42 refs · 11 cites · 200 2-hop

Click a neighbor to re-center.

Edges

- center cites neighbor (refs)
- neighbor cites center (cites)
- 2-hop link

Nodes

- center (thick outline)
- 1-hop neighbor
- 2-hop sample

Fill colour = community. Radius = $1 + \log_2(\text{weight})$ for 2-hop nodes.

